

Project Report

Project Title

AI-Driven Buyer Segmentation & Investment Behavior Analysis for Parcl Real Estate Platform

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Introduction

I completed the project **AI-Driven Buyer Segmentation & Investment Behavior Analysis** for the Parcl real estate platform. The objective of this project was to understand different types of property buyers and analyze their investment behavior using data analytics techniques.

Problem Statement

Parcl lacked a data-driven understanding of buyer categories, investment motivations, geographic investment behavior, and financing patterns. Because of this, marketing campaigns and property recommendations were not sufficiently targeted.

Dataset Used

The project used two datasets:

- **clients.csv** containing buyer demographics, financing information, and satisfaction details.
- **properties.csv** containing property transaction and investment information.

The datasets were connected using the **client_id / client_ref** relationship.

Tools Used

- Microsoft Excel 2010
- Microsoft Power BI Desktop
- Streamlit Cloud
- GitHub for project hosting

Methodology

I performed the following steps:

1. Data cleaning and removal of duplicate records.
2. Handling missing values.
3. Creation of age and buyer-segment variables in Excel.
4. Buyer segmentation into:

- Global Investors
 - First-Time Buyers
 - Corporate Buyers
 - Luxury Investors
5. KPI analysis using PivotTables.
 6. Interactive dashboard creation in Power BI.
 7. Online deployment using Streamlit Cloud.

Key Findings

The analysis revealed that investment behavior differs across regions and buyer types. First-time buyers showed higher loan dependency, corporate buyers represented institutional demand, and investment-focused buyers were concentrated in specific geographic regions.

Business Recommendations

- Target global investors with investment-oriented campaigns.
- Provide financing assistance for first-time buyers.
- Offer dedicated relationship management for corporate buyers.
- Promote premium listings to luxury investors.

Conclusion

This project demonstrated how Excel and Power BI can be used to build a practical buyer intelligence solution without advanced coding. The dashboard supports better customer targeting, marketing efficiency, and data-driven real estate investment decisions.

Thank you.